Thermo Fisher SCIENTIFIC

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

Creation Date 26-Nov-2009 **Revision Date** 29-Sep-2023 **Revision Number** 10

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description: **Ammonia, 0.5M solution in 1,4-dioxane**
Cat No. : **368380000; 368380010; 368381000**

Unique Formula Identifier (UFI) **CN1R-SU0M-3W0X-UJJ3**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Sector of use	SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites
Product category	PC21 - Laboratory chemicals
Process categories	PROC15 - Use as a laboratory reagent
Environmental release category	ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company

EU entity/business name
Thermo Fisher Scientific
Janssen Pharmaceuticalaan 3a, 2440 Geel, Belgium

UK entity/business name
Fisher Scientific UK
Bishop Meadow Road,
Loughborough, Leicestershire LE11 5RG, United Kingdom

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
e-mail - infoch@thermofisher.com

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US:**001-201-796-7100 / **Europe:** +32 14 57 52 99
CHEMTREC Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

Poison Centre - Emergency information services

Ireland : National Poisons Information Centre (NPIC) -
01 809 2166 (8am-10pm, 7 days a week)
Malta : +356 2395 2000
Cyprus : +357 2240 5611

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Flammable liquids	Category 2 (H225)

Health hazards

Serious Eye Damage/Eye Irritation	Category 2 (H319)
Carcinogenicity	Category 1B (H350)
Specific target organ toxicity - (single exposure)	Category 3 (H335)

Environmental hazards
Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements

None required

Signal Word **Danger**

H225 - Highly flammable liquid and vapor
H319 - Causes serious eye irritation
H335 - May cause respiratory irritation
H350 - May cause cancer
EUH019 - May form explosive peroxides
EUH066 - Repeated exposure may cause skin dryness or cracking

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
P312 - Call a POISON CENTER or doctor if you feel unwell
P280 - Wear protective gloves/protective clothing/eye protection/face protection

Additional EU labelling
Restricted to professional users

2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Contains a known or suspected endocrine disruptor
Included in the list established in accordance with Article 59(1) for having endocrine disrupting properties
Toxic to terrestrial vertebrates

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
1,4-Dioxane	123-91-1	EEC No. 204-661-8	> 99	Flam. Liq. 2 (H225) Eye Irrit. 2 (H319) STOT SE 3 (H335) Carc. 1B (H350) EUH019 EUH066
Ammonia	7664-41-7	EEC No. 231-635-3	< 1	Flam. Gas 2 (H221) Skin Corr. 1B (H314) Acute Tox. 3 (H331) Aquatic Acute 1 (H400) Aquatic Chronic 2 (H411) (EUH071)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Ammonia	STOT SE 3 : C ≥ 5 %	1	-

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice If symptoms persist, call a physician.

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

Ingestion Clean mouth with water and drink afterwards plenty of water.

Inhalation Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

Self-Protection of the First Aider Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically. Symptoms may be delayed.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
Water spray, carbon dioxide (CO2), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

Extinguishing media which must not be used for safety reasons
No information available.

5.2. Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

Hazardous Combustion Products
Carbon monoxide (CO), Carbon dioxide (CO_2).

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Ensure adequate ventilation. If peroxide formation is suspected, do not open or move container. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Refrigerator/flammables. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals.

Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK) (Germany) Class 3

Switzerland - Storage of hazardous substances Storage class - SC 3
https://www.kvu.ch/de/themen/stoffe-und-produkte
https://www.kvu.ch/fr/themes/substances-et-produits
https://www.kvu.ch/it/temi/sostanze-e-prodotti

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits
List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE -** 2010 Code of Practice for the Safety, Health and Safety and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
1,4-Dioxane	TWA: 20 ppm (8h) TWA: 73 mg/m³ (8h)	STEL: 60 ppm 15 min STEL: 219 mg/m³ 15 min TWA: 20 ppm 8 hr TWA: 73 mg/m³ 8 hr Skin	TWA / VME: 20 ppm (8 heures). restrictive limit TWA / VME: 73 mg/m³ (8 heures). restrictive limit STEL / VLCT: 40 ppm. restrictive limit STEL / VLCT: 140 mg/m³. restrictive limit	TWA: 20 ppm 8 uren TWA: 73 mg/m³ 8 uren Huid	TWA / VLA-ED: 20 ppm (8 horas) TWA / VLA-ED: 73 mg/m³ (8 horas)
Ammonia	TWA: 20 ppm (8h) TWA: 14 mg/m³ (8h) STEL: 50 ppm (15min) STEL: 36 mg/m³ (15min)	STEL: 35 ppm 15 min STEL: 25 mg/m³ 15 min TWA: 25 ppm 8 hr TWA: 18 mg/m³ 8 hr	TWA / VME: 10 ppm (8 heures). restrictive limit TWA / VME: 7 mg/m³ (8 heures). restrictive limit STEL / VLCT: 20 ppm. restrictive limit STEL / VLCT: 14 mg/m³. restrictive limit	TWA: 20 ppm 8 uren TWA: 14 mg/m³ 8 uren STEL: 50 ppm 15 minuten STEL: 36 mg/m³ 15 minuten	STEL / VLA-EC: 50 ppm (15 minutos). STEL / VLA-EC: 36 mg/m³ (15 minutos). TWA / VLA-ED: 20 ppm (8 horas) TWA / VLA-ED: 14 mg/m³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
1,4-Dioxane	Pelle	TWA: 20 ppm (8 Stunden). AGW - exposure factor 2 TWA: 73 mg/m³ (8 Stunden). AGW - exposure factor 2 TWA: 10 ppm (8 Stunden). MAK	TWA: 20 ppm 8 horas TWA: 73 mg/m³ 8 horas Pele	TWA: 20 mg/m³ 8 uren	TWA: 10 ppm 8 tunteina TWA: 36 mg/m³ 8 tunteina STEL: 40 ppm 15 minuutteina STEL: 150 mg/m³ 15 minuutteina Iho

		TWA: 37 mg/m^3 (8 Stunden). MAK Höhepunkt: 20 ppm Höhepunkt: 74 mg/m^3 Haut			
Ammonia	TWA: 20 ppm 8 ore. Time Weighted Average TWA: 14 mg/m^3 8 ore. Time Weighted Average STEL: 50 ppm 15 minuti. Short-term STEL: 36 mg/m^3 15 minuti. Short-term	TWA: 20 ppm (8 Stunden). AGW - exposure factor 2 TWA: 14 mg/m^3 (8 Stunden). AGW - exposure factor 2 TWA: 20 ppm (8 Stunden). MAK TWA: 14 mg/m^3 (8 Stunden). MAK Höhepunkt: 40 ppm Höhepunkt: 28 mg/m^3	STEL: 50 ppm 15 minutos STEL: 36 mg/m^3 15 minutos TWA: 20 ppm 8 horas TWA: 14 mg/m^3 8 horas	STEL: 36 mg/m^3 15 minuten TWA: 14 mg/m^3 8 uren	TWA: 20 ppm 8 tunteina TWA: 14 mg/m^3 8 tunteina STEL: 50 ppm 15 minuutteina STEL: 36 mg/m^3 15 minuutteina

Component	Austria	Denmark	Switzerland	Poland	Norway
1,4-Dioxane	Haut MAK-KZGW: 40 ppm 15 Minuten MAK-KZGW: 146 mg/m^3 15 Minuten MAK-TMW: 20 ppm 8 Stunden MAK-TMW: 73 mg/m^3 8 Stunden	TWA: 10 ppm 8 timer TWA: 36 mg/m^3 8 timer STEL: 20 ppm 15 minutter STEL: 72 mg/m^3 15 minutter Hud	Haut/Peau STEL: 40 ppm 15 Minuten STEL: 144 mg/m^3 15 Minuten TWA: 20 ppm 8 Stunden TWA: 72 mg/m^3 8 Stunden	TWA: 50 mg/m^3 8 godzinach	TWA: 5 ppm 8 timer TWA: 18 mg/m^3 8 timer STEL: 10 ppm 15 minutter. value from the regulation STEL: 36 mg/m^3 15 minutter. value from the regulation Hud
Ammonia	MAK-KZGW: 50 ppm 15 Minuten MAK-KZGW: 36 mg/m^3 15 Minuten MAK-TMW: 20 ppm 8 Stunden MAK-TMW: 14 mg/m^3 8 Stunden	TWA: 20 ppm 8 timer TWA: 14 mg/m^3 8 timer STEL: 36 mg/m^3 15 minutter STEL: 50 ppm 15 minutter	STEL: 40 ppm 15 Minuten STEL: 28 mg/m^3 15 Minuten TWA: 20 ppm 8 Stunden TWA: 14 mg/m^3 8 Stunden	STEL: 28 mg/m^3 15 minutach TWA: 14 mg/m^3 8 godzinach	TWA: 15 ppm 8 timer TWA: 11 mg/m^3 8 timer TWA: 20 ppm 8 timer STEL: 50 ppm 15 minutter. value from the regulation STEL: 36 mg/m^3 15 minutter. value from the regulation STEL: 30 ppm 15 minutter. a transitional norm valid 2013-2024, applies to farmers at livestock production buildings constructed before 2002;value calculated

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
1,4-Dioxane	TWA: 20 ppm TWA: 73 mg/m^3	TWA-GVI: 20 ppm 8 satima. TWA-GVI: 73 mg/m^3 8 satima.	TWA: 20 ppm 8 hr. technical grade TWA: 73 mg/m^3 8 hr. technical grade STEL: 60 ppm 15 min STEL: 219 mg/m^3 15 min Skin	TWA: 73 mg/m^3 TWA: 20 ppm	TWA: 70 mg/m^3 8 hodinách. Potential for cutaneous absorption Ceiling: 140 mg/m^3
Ammonia	TWA: 14.0 mg/m^3 TWA: 20 ppm STEL : 50 ppm STEL : 36.0 mg/m^3	TWA-GVI: 20 ppm 8 satima. TWA-GVI: 14 mg/m^3 8 satima. STEL-KGVI: 50 ppm 15 minutama. STEL-KGVI: 36 mg/m^3 15 minutama.	TWA: 20 ppm 8 hr. anhydrous TWA: 14 mg/m^3 8 hr. anhydrous STEL: 50 ppm 15 min STEL: 36 mg/m^3 15 min	STEL: 50 ppm STEL: 36 mg/m^3 TWA: 20 ppm TWA: 14 mg/m^3	TWA: 14 mg/m^3 8 hodinách. Ceiling: 36 mg/m^3

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
1,4-Dioxane	TWA: 20 ppm 8 tundides. TWA: 73 mg/m^3 8 tundides.	TWA: 73 mg/m^3 8 hr TWA: 20 ppm 8 hr	TWA: 20 ppm TWA: 73 mg/m^3	TWA: 73 mg/m^3 8 órában. AK lehetséges borön keresztüli felszívódás	TWA: 20 ppm 8 klukkustundum. TWA: 73 mg/m^3 8 klukkustundum.

					Skin notation Ceiling: 40 ppm Ceiling: 146 mg/m³
Ammonia	TWA: 20 ppm 8 tundides. TWA: 14 mg/m³ 8 tundides. STEL: 50 ppm 15 minutites. STEL: 36 mg/m³ 15 minutites.		STEL: 50 ppm STEL: 35 mg/m³ TWA: 50 ppm TWA: 35 mg/m³	STEL: 36 mg/m³ 15 percekben. CK TWA: 14 mg/m³ 8 órában. AK	STEL: 50 ppm 5 minutes STEL: 36 mg/m³ 5 minutes TWA: 20 ppm 8 klukkustundum. TWA: 14 mg/m³ 8 klukkustundum. Skin notation

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
1,4-Dioxane	TWA: 5.5 ppm TWA: 20 mg/m³	TWA: 10 ppm IPRD TWA: 35 mg/m³ IPRD STEL: 25 ppm STEL: 90 mg/m³	TWA: 73 mg/m³ 8 Stunden TWA: 20 ppm 8 Stunden	TWA: 73 mg/m³ TWA: 20 ppm	Skin notation TWA: 20 ppm 8 ore TWA: 73 mg/m³ 8 ore
Ammonia	STEL: 50 ppm STEL: 36 mg/m³ TWA: 20 ppm TWA: 14 mg/m³	TWA: 20 ppm IPRD TWA: 14 mg/m³ IPRD STEL: 50 ppm STEL: 36 mg/m³	TWA: 20 ppm 8 Stunden TWA: 14 mg/m³ 8 Stunden STEL: 50 ppm 15 Minuten STEL: 36 mg/m³ 15 Minuten	TWA: 20 ppm TWA: 14 mg/m³ STEL: 50 ppm 15 minuti STEL: 36 mg/m³ 15 minuti	TWA: 20 ppm 8 ore TWA: 14 mg/m³ 8 ore STEL: 50 ppm 15 minute STEL: 36 mg/m³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
1,4-Dioxane	Skin notation MAC: 10 mg/m³	Ceiling: 146 mg/m³ TWA: 20 ppm TWA: 73 mg/m³	TWA: 20 ppm 8 urah TWA: 73 mg/m³ 8 urah Koža STEL: 146 mg/m³ 15 minutah STEL: 40 ppm 15 minutah	Indicative STEL: 25 ppm 15 minuter Indicative STEL: 90 mg/m³ 15 minuter TLV: 10 ppm 8 timmar. NGV TLV: 35 mg/m³ 8 timmar. NGV	TWA: 20 ppm 8 saat TWA: 73 mg/m³ 8 saat
Ammonia	MAC: 20 mg/m³	Ceiling: 36 mg/m³ TWA: 20 ppm TWA: 14 mg/m³	TWA: 20 ppm 8 urah TWA: 14 mg/m³ 8 urah STEL: 50 ppm 15 minutah anhydrous STEL: 36 mg/m³ 15 minutah anhydrous	Binding STEL: 50 ppm 15 minuter Binding STEL: 36 mg/m³ 15 minuter TLV: 20 ppm 8 timmar. NGV TLV: 14 mg/m³ 8 timmar. NGV	TWA: 20 ppm 8 saat TWA: 14 mg/m³ 8 saat STEL: 50 ppm 15 dakika STEL: 36 mg/m³ 15 dakika

Biological limit values
List source(s):

Component	European Union	United Kingdom	France	Spain	Germany
1,4-Dioxane					2-Hydroxyethoxyacetic acid: 200 mg/g Creatinine urine (end of shift)

Monitoring methods
BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.
MDHS70 General methods for sampling airborne gases and vapours
MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography
MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)
See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Ammonia 7664-41-7 (< 1)		DNEL = 6.8mg/kg bw/day		DNEL = 6.8mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Ammonia 7664-41-7 (< 1)	DNEL = 36mg/m³	DNEL = 47.6mg/m³	DNEL = 14mg/m³	DNEL = 47.6mg/m³

Predicted No Effect Concentration (PNEC)
See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Ammonia 7664-41-7 (< 1)	PNEC = 0.0011mg/L		PNEC = 0.0068mg/L		

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Ammonia 7664-41-7 (< 1)	PNEC = 0.0011mg/L				

8.2. Exposure controls

Engineering Measures
Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure adequate ventilation, especially in confined areas.
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection Goggles (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Butyl rubber	> 480 minutes	0.7 mm	EN 374	As tested under EN374-3 Determination of
Viton (R)	> 480 minutes	0.7 mm		Resistance to Permeation by Chemicals
Butyl rubber	< 200 minutes	0.35 mm		

Skin and body protection Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits

	are exceeded or if irritation or other symptoms are experienced **Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387
Small scale/Laboratory use	Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. **Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141 When RPE is used a face piece Fit Test should be conducted
Environmental exposure controls	Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance	Colorless	
Odor	Ammonia	
Odor Threshold	No data available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	No information available	
Flammability (liquid)	Highly flammable	On basis of test data
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Flash Point	11 °C / 51.8 °F	**Method -** No information available
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	No information available	
Viscosity	No data available	
Water Solubility	Soluble	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	**log Pow**	
1,4-Dioxane	-0.42	
Vapor Pressure	No data available	
Density / Specific Gravity	1.023	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Explosive Properties Vapors may form explosive mixtures with air

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under recommended storage conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization Hazardous polymerization does not occur.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.

10.5. Incompatible materials

Strong oxidizing agents. Halogens. Strong acids.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO_2).

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;
Oral Based on available data, the classification criteria are not met
Dermal Based on available data, the classification criteria are not met
Inhalation Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
1,4-Dioxane	5170 mg/kg (Rat) 4200 mg/kg (Rat)	LD50 = 7600 mg/kg (Rabbit)	48.5 mg/L (Rat) 4 h
Ammonia	LD50 = 350 mg/kg (Rat)	-	LC50 = 9850 mg/m³ (Rat) 1 h LC50 = 13770 mg/m³ (Rat) 1 h

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;
Respiratory Based on available data, the classification criteria are not met
Skin Based on available data, the classification criteria are not met

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Category 1B

The table below indicates whether each agency has listed any ingredient as a carcinogen

Component	EU	UK	Germany	IARC
1,4-Dioxane	Carc Cat. 1B			Group 2B

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Category 3

Results / Target organs Respiratory system.

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

Symptoms / effects,both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

11.2. Information on other hazards

Endocrine Disrupting Properties .
Assess endocrine disrupting properties for human health Substance identified as having endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity
Ecotoxicity effects Contains a substance which is:. Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae
1,4-Dioxane	LC50: = 9850 mg/L, 96h (Pimephales promelas) LC50: 10306 - 14742 mg/L, 96h static (Pimephales promelas) LC50: = 9850 mg/L, 96h flow-through (Pimephales promelas) LC50: > 10000 mg/L, 96h semi-static (Lepomis macrochirus) LC50: > 10000 mg/L, 96h static (Lepomis macrochirus)	EC50 = 163 mg/L 48h	
Ammonia	LC50: 0.26 - 4.6 mg/L, 96h (Lepomis macrochirus) LC50: = 1.17 mg/L, 96h flow-through (Lepomis macrochirus) LC50: 0.73 - 2.35 mg/L, 96h (Pimephales promelas) LC50: = 5.9 mg/L, 96h static (Pimephales promelas) LC50: > 1.5 mg/L, 96h (Poecilia reticulata) LC50: = 1.19 mg/L, 96h static (Poecilia reticulata) LC50: = 0.44 mg/L, 96h (Cyprinus carpio)	EC50 = 25.4 mg/L, 48h (Daphnia magna) NOEC = 0.79 mg/L (Daphnia magna)	

Component	Microtox	M-Factor
1,4-Dioxane	EC50 = 610 mg/L 5 min EC50 = 668 mg/L 15 min EC50 = 733 mg/L 30 min	
Ammonia	EC50 = 2.0 mg/L 5 min	1

12.2. Persistence and degradability

Persistence Persistence is unlikely.

Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
1,4-Dioxane	-0.42	0.3 - 0.7 dimensionless

12.4. Mobility in soil The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

12.5. Results of PBT and vPvB assessment Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

12.6. Endocrine disrupting properties

Endocrine Disruptor Information

Assess endocrine disrupting properties for the environment Substance identified as having endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

12.7. Other adverse effects

Persistent Organic Pollutant This product does not contain any known or suspected substance

Ozone Depletion Potential This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

European Waste Catalogue (EWC) According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations.

Switzerland - Waste Ordinance Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
https://www.fedlex.admin.ch/eli/cc/2015/891/en

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number UN1993

14.2. UN proper shipping name Flammable liquid, n.o.s.
Technical Shipping Name 1,4-Dioxane
14.3. Transport hazard class(es) 3
14.4. Packing group II

ADR

14.1. UN number UN1993
14.2. UN proper shipping name Flammable liquid, n.o.s.
Technical Shipping Name 1,4-Dioxane
14.3. Transport hazard class(es) 3
14.4. Packing group II

IATA

14.1. UN number UN1993
14.2. UN proper shipping name Flammable liquid, n.o.s.
Technical Shipping Name 1,4-Dioxane
14.3. Transport hazard class(es) 3
14.4. Packing group II

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories
Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
1,4-Dioxane	123-91-1	204-661-8	-	-	X	X	KE-10463	X	X
Ammonia	7664-41-7	231-635-3	-	-	X	X	KE-01625	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
1,4-Dioxane	123-91-1	X	ACTIVE	X	-	X	X	X
Ammonia	7664-41-7	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
1,4-Dioxane	123-91-1	-	Use restricted. See item 75. (see link for restriction details)	SVHC Candidate list - 204-661-8 - Carcinogenic (Article 57a)

			Use restricted. See item 28. (see link for restriction details)	Equivalent level of concern having probable serious effects to the environment (Article 57f - environment) Equivalent level of concern having probable serious effects to human health (Article 57f - human health)
Ammonia	7664-41-7	-	Use restricted. See item 75. (see link for restriction details)	-

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

REACH links
https://echa.europa.eu/substances-restricted-under-reach
https://echa.europa.eu/candidate-list-table
https://echa.europa.eu/authorisation-list

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
1,4-Dioxane	123-91-1	Not applicable	Not applicable
Ammonia	7664-41-7	50 tonne	200 tonne

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals
Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?
Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .
Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values
Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification Water endangering class = 3 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
1,4-Dioxane	WGK2	Class I : 20 mg/m³ (Massenkonzentration)
Ammonia	WGK2	

Component	France - INRS (Tables of occupational diseases)
1,4-Dioxane	Tableaux des maladies professionnelles (TMP) - RG 84

Swiss Regulations
Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).
Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
1,4-Dioxane 123-91-1 (> 99)		Group I	

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H319 - Causes serious eye irritation
H335 - May cause respiratory irritation
H350 - May cause cancer
EUH019 - May form explosive peroxides
EUH066 - Repeated exposure may cause skin dryness or cracking
H221 - Flammable gas
H225 - Highly flammable liquid and vapor
H314 - Causes severe skin burns and eye damage
H331 - Toxic if inhaled
H400 - Very toxic to aquatic life
H411 - Toxic to aquatic life with long lasting effects
EUH071 - Corrosive to the respiratory tract

Legend

CAS - Chemical Abstracts Service
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
ENCS - Japanese Existing and New Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC - (volatile organic compound)

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards	On basis of test data
Health Hazards	Calculation method
Environmental hazards	Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.
First aid for chemical exposure, including the use of eye wash and safety showers.
Chemical incident response training.
Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Creation Date	26-Nov-2009
Revision Date	29-Sep-2023
Revision Summary	SDS sections updated.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

End of Safety Data Sheet